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$$A = \frac{\pi}{4} d^2$$

$$\rho = \frac{\pi d^2 R}{4 l} = \frac{\pi (6.3 \times 10^{-3})^2 (939)}{4 \times 2.710} = 0.010801082$$

C1

$$\frac{\Delta \rho}{\rho} = \frac{2 \Delta d}{d} + \frac{\Delta R}{R} + \frac{\Delta l}{l} = \frac{2(0.05)}{6.30} + \frac{3}{939} + \frac{0.01}{2.71}$$

C1

$$\Delta \rho = (0.02276)(0.01080) = 2.456 \times 10^{-4} \text{ or } 2.5 \times 10^{-4}$$

$$= 0.0002 \text{ (1s.f) or } 0.0003 \text{ (1s.f)}$$

M1

$$\rho \pm \Delta \rho = (0.0108 \pm 0.0002) \Omega \text{ m}$$

A1

$$\text{Or } (108 \pm 2) \times 10^{-4} \Omega$$

* Both 0.0002 and 0.0003 are acceptable

Alternative method (max and min)

$$\text{Max } \rho = \frac{(939+3)\pi (6.35 \times 10^{-3})^2}{4(2.69)} = 0.01109$$

$$\text{Min } \rho = \frac{(939-3)\pi (6.25 \times 10^{-3})^2}{4(2.72)} = 0.010557$$

$$\Delta \rho = \frac{0.01109 - 0.010557}{2} = 0.000246$$